Torsional friction in planar contact: numerical and simulation studies

Experimental report

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Abstract

This report collects the numerical and simulation evidence for the theory of the companion document "Torsional friction in planar contact" (below: *the theory*; result numbers such as Theorem 4.2 refer to it). The theory bounds the frictional capacity of a planar contact patch over all normal-pressure fields compatible with a measurement, with torques about a fixed point, and turns the bounds into a certified and an upper load factor. We test it in two steps. First, on a population of analytic contact-mechanics pressure fields, we measure how tight the bounds and the certificate are, how they depend on taxel resolution and on measuring the centre of pressure (CoP), and how they compare with plug-in estimates: the gap halves with each dyadic refinement, and measuring the CoP in addition to the taxel forces narrows the bounds by about 30% for mixed shear–torsion loads. Second, in Drake hydroelastic simulations of a bar held between two compliant pads, we close the loop: a grip controller squeezes just enough to keep the demanded load certified. Certified policies never slipped without warning in 2160 episodes, at 1.16–1.18 times the squeeze of an oracle with 16×16 taxels, while nominal estimators tuned on one set of contacts slipped when the contact geometry changed. A sensing-delay study shows the limit of a static certificate: when the load is measured rather than known in advance, the load growth during the loop lag requires a separate temporal margin. The report ends with the planned robust extension and hardware validation, whose outcomes are open.

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1 Scientific questions and approach

Story. A tactile sensor reports how hard each region of a contact is pressed, never the exact pressure field. The tangential force and torsion that a contact can hold depend on that field: pressure near the rim resists twisting, pressure concentrated at one point does not. The theory bounds the admissible tangential wrench set $\mathcal{X}(\nu_N)$, with the torque about a fixed point (Proposition 3.2), over all normal-force measures ν_N compatible with a measurement y: the *guaranteed* (inner) set $\mathcal{X}^-(y)$ and the *possible* (outer) set $\mathcal{X}^+(y)$, which is convex (Definition 3.8, Proposition 3.10). Three measurement models are considered: $\mathcal{M}_0$ (total normal force and CoP, i.e. an F/T sensor), $\mathcal{M}_F$ (taxel forces) and $\mathcal{M}_T$ (taxel forces and CoP). For control, the certified load factor (Proposition 9.4) gives the largest scaling of a load that every compatible contact carries (two SOCPs), and the upper load factor the scaling beyond which no compatible contact carries it (one SOCP): a load is surely held, surely not held, or undecided.

Questions. The studies answer six questions.

Q1 Correctness. Do the finite-dimensional reductions and the certificate agree with brute-force computations and with the capacity of simulated contacts?

Q2 Tightness and resolution. How far are the guaranteed and possible capacities from the truth, and how does the gap shrink with the taxel resolution (Propositions 8.1–8.3)?

Q3 Value of the CoP. How much does measuring the CoP ($\mathcal{M}_T$ vs $\mathcal{M}_F$) tighten the bounds?

Q4 Plug-in estimation. Is it safe to treat the measured taxel forces as an exact pressure field?

Q5 Closed loop. Does a grip controller that keeps the load certified avoid slip, at what squeeze, and how does it compare with nominal estimators with tuned margins?

Q6 Temporal validity. The certificate is static. How much sensing delay and load rate does the closed loop tolerate?

Relation to the literature. The coarse case $\mathcal{M}_0$ is prior art: the bounds over pressures with a known total force and CoP are those of Peshkin and Sanderson (1988), the extremal loads are the tripods noted by Alexander and Maddocks (1993), the construction is the moment envelope of Huang et al. (2017), and the vertex bound goes back to Madansky (1959) and Gassmann and Ziemba (1986); the support function is the Coulomb dissipation (Goyal et al., 1991; Alexander and Maddocks, 1993). The taxel cases generalize partition bounds with known cell means (Gassmann and Ziemba, 1986; Birge and Wets, 1983) to unknown cell means, and stay exact although taxel partitions violate the nesting condition of Wiesemann et al. (2014). Tactile capacity estimation, by contrast, uses one measured or assumed pressure field (Bicchi et al., 1988; Howe and Cutkosky, 1996; Costanzo et al., 2020; Arslan Waltersson and Karayiannidis, 2025). The studies below therefore focus on what is new: the taxel bounds, the certified load factor, and their use in closed loop. The full audit is in `research/literature_review.md`.

Study	Section	Questions	Data	Code
Verification	4.1	Q1	analytic fields	`tests/`, `examples/01,02`
F1 wrench sets	4.3	Q2, Q3	one partial contact	`examples/07`
F2 resolution	4.4	Q2, Q3	field population	`examples/03,08`
F3 plug-in	4.5	Q4	field population	`examples/09`
Load factor	4.6	Q2–Q4	field population	`examples/10`
S1 contact read-out	5.2	Q1	Drake	`sim/smoke_contact.py`
S2 break-away	5.3	Q1	Drake	`sim/bar_breakaway.py`, `sweep_breakaway.py`
S3 closed-loop grip	5.4	Q5	Drake, 3780 episodes	`sim/grip_study.py`
S4 sensing delay	5.5	Q6	Drake, 5400 episodes	`sim/delay_study.py`

Table 1: Studies, the questions they address, and their code (repository paths).

What is and is not tested. The numerical studies verify and quantify the theory on pressure fields from contact mechanics; they are not independent validation. The Drake studies test the decision pipeline (sensing → certificate → command → outcome) and the effects the theory does not model: compliant pads, actuator lag, load dynamics, sensing delay, the split of the load between two pads. They do *not* test Coulomb friction itself, which Drake implements (Section 5). Whether the Coulomb patch model holds on a real gel is the purpose of the planned hardware study (Section 7.2).

2 Quantities, ground truth and metrics

Support values and load factors. Two kinds of quantities are reported.

- *Support values* $h_{\mathcal{X}(\nu_N)}(q)$ for a twist $q = (a, b)$, with their bounds $\underline{h}(q; y, \mathcal{M}) \le h \le \overline{h}(q; y, \mathcal{M})$ (Definition 3.8). A twist is a physical object, written about a stated point: in F2 and F3, $q = (0, 0, 1)$ is the unit rotation about the true CoP of each field, a fixed twist for every compatible measure. Support values describe the sets; F1–F3 use them.
- *Load factors* $s^\star(\nu_N; w) = \max\{s \ge 0 : sw \in \mathcal{X}(\nu_N)\}$ (Definition 9.1), the capacity along a load w given about a fixed point o (moved to the origin by Varignon's formula), with the certified (lower) and upper load factors $\underline{s}(w; y, \mathcal{M}) \le s^\star \le \overline{s}(w; y, \mathcal{M})$. For a pure torque $w = (0, 0, 1)$ this is the maximal torque $\inf_a h((a, 1))$, which is at most $h((0, 0, 1))$. Load factors are what a controller needs; the load-factor study and S2–S4 use them.

Ground truth. For an analytic field sampled on a grid, the true support value is the quadrature $\mu \sum_j p_j A_j \, g(x_j; q)$ with the integrand of Lemma 3.12. The population studies use a 129×129 support grid on the unit pad (4 grid spacings per taxel at 32×32). In Drake, each contact-surface face is a Coulomb point contact at its centroid with force $A_f p_f$ (Masterjohn et al., 2022), so the true capacity of a simulated contact is exactly the support function of the discrete measure $\sum_f A_f p_f \, \delta_{x_f}$, and its load factor is computed exactly by `pfb.certify.load_factor_true`.

Ratios. Tightness is reported as bound / truth: below 1 for a sound lower bound, above 1 for an upper bound. An estimate above the truth is *unsafe*: it predicts that a load is held when it may slip.

Closed-loop outcomes (S3, S4). Each episode ends in one of three outcomes:

- **success:** no slip and no flag;
- **infeasible (flagged):** before any slip, the requested squeeze exceeded the force budget, so by its own estimate the policy cannot carry the load; a real controller would stop or replan;

- **silent slip:** slip above 0.2 mm at the pad faces without an earlier flag.

Effort is the mean squeeze of an episode relative to the oracle on the same seed. Medians over successes are biased toward easy seeds when the success rate is low, so the main comparisons use *matched seeds*: the seeds where every compared policy succeeds.

Implementation. The library `pfb` implements the exact forms of the theory, with torques and twists about a fixed point: the $\mathcal{M}_F$ bounds in closed form (Theorem 6.1), the $\mathcal{M}_T$ lower bound as an SOCP in the local first moments (Theorem 7.1), the $\mathcal{M}_T$ and $\mathcal{M}_0$ upper bounds as corner LPs (Theorem 7.2, Corollary 4.5), the certified load factor (two SOCPs after the substitution $u_i = bc_i$) and the upper load factor (one SOCP; Proposition 9.5). Conic programs are solved with CVXPY/Clarabel and the brute-force grid LPs with HiGHS. Lengths are in pad sides for the analytic studies, and $\mu = 1$ there ($\mu = 0.8$ in Drake).

3 Methods compared

All methods receive the same measurement and return an estimate of the capacity (support value or load factor). In the closed loop they plug into the same controller (Section 5.4).

Oracle.
The true capacity of the current contact, computed from Drake's contact faces. It is not available in practice; it defines the minimal squeeze and calibrates the dynamic margin of the loop.

Certified $\mathcal{M}_T$ ($k \times k$ taxels).
The certified load factor of Proposition 9.4 from $k \times k$ taxel forces and the CoP on each pad, summed over the two pads (Proposition 9.6). $k \in \{2, 4, 8, 16\}$ on a 32×32 mm footprint per pad.

Certified $\mathcal{M}_F$ (16×16).
The same without the CoP.

Certified, one pad sensed.
$\mathcal{M}_T$ 16×16 on one pad; the other pad is unsensed except for its normal force, and contributes its worst case over the pad, $\mu f_N \min_{x \in P_2} \|a + bJ(x - o)\|$. This is the hardware configuration (one GelSight finger).

$\mathcal{M}_0$ bounds.
Total force and CoP only (Section 4); they are the bounds of Peshkin and Sanderson (1988) and Huang et al. (2017) for capacity. Their guaranteed torsion about the CoP is zero (Corollary 4.3).

Plug-in ($k \times k$).
Each taxel force spread uniformly over its taxel, and the capacity of that field taken as exact (`pfb.plugin_uniform`). This is the classical use of tactile-array pressure (Bicchi et al., 1988; Howe and Cutkosky, 1996).

F/T + contact mask.
The measured total force spread uniformly over the measured contact area (the contact faces, weighted by area), then its exact load factor: uniform pressure over the contact, the classical limit-surface assumption (Howe and Cutkosky, 1996), with the contact geometry assumed known (e.g. from a vision-based tactile image).

Uniform over the pad (F/T only).
The total force spread over the whole sensor footprint: what an F/T sensor plus the pad geometry suggests without contact sensing.

Margins. Every method is used with a multiplicative margin m: the controller targets $D(1+m)$ instead of the demand D. For the oracle, m covers the loop dynamics only; the smallest margin at which the oracle never slipped, $m_{\mathrm{dyn}} = 0.05$, is the *dynamic margin* of the loop. Certified methods use m_{dyn}; nominal methods (plug-in, mask, uniform) need extra margin for their estimation error, which is tuned on a training set and tested on held-out data (Section 5.4).

Not implemented. The parametric axisymmetric model of Costanzo et al. (2020) (calibrated $\tau_{\max} \propto f_n^{\gamma+1}$), the ellipsoidal limit surface (Howe and Cutkosky, 1996; Xydas and Kao, 1999) and probing-based radius identification (Arslan Waltersson and Karayiannidis, 2025) need a calibrated physical pad; the first is planned as a hardware baseline (Section 7.2).

4 Numerical studies on contact-mechanics pressure fields

4.1 Verification

Question. Do the reductions of the theory compute what they claim (Q1)?

Protocol.

- The exact forms (`pfb.reduced`) are compared with brute-force grid LPs over all measures on a fine grid with the same measurement constraints (`pfb.bounds`), on twists over the whole sphere about the pad centre; the script (`examples/02_bounds_vs_truth.py`) also checks the orderings of Proposition 8.1.
- The script `examples/01_counterexample.py` evaluates the non-identifiability example (Example 4.7).
- Unit tests (`pixi run test`), one per statement of the theory: the change of reference point, linearity, convexity of the possible set, the closed forms of $\mathcal{M}_0$ and $\mathcal{M}_F$, monotonicity along dyadic grid chains, the gap bounds and the first-order rate (Theorem 8.4), the certified load factor against finite direction grids and a brute-force minimum over point loads, its tightness (Proposition 9.4) for one and two pads, the upper load factor (primal and dual programs, attained by a corner measure), the analytic $\mathcal{M}_0$ case (no certified torsion about the CoP, Corollary 4.3), and the field generators.

Results. On one field with 4×4 taxels and twists over the whole sphere, all orderings hold: $\max(\underline{h}_{\mathcal{M}_0}, \underline{h}_{\mathcal{M}_F}) \le \underline{h}_{\mathcal{M}_T} \le h \le \overline{h}_{\mathcal{M}_T} \le \min(\overline{h}_{\mathcal{M}_0}, \overline{h}_{\mathcal{M}_F})$ (Proposition 8.1). The brute-force grid LPs lie inside the exact bounds, with differences of $7 \cdot 10^{-11}$ ($\mathcal{M}_0$), $1.6 \cdot 10^{-3}$ ($\mathcal{M}_T$) and $5.0 \cdot 10^{-3}$ ($\mathcal{M}_F$), the discretization error of the grid (closed-cell model); the mean bound widths are 0.128 ($\mathcal{M}_0$), 0.190 ($\mathcal{M}_F$) and 0.048 ($\mathcal{M}_T$). Example 4.7: two measures with the same total force and CoP have supports 0 and 0.5 ($= \mu f_N L/2$) for the rotation about the CoP, and the $\mathcal{M}_0$ bounds are $[0, 0.5]$. The 113 unit tests pass.

Interpretation. The exact forms are exact where the theory says so, and both load factors are attained by compatible measures. A direction grid, by contrast, *over*-certifies: minimizing $\underline{h}$ over finitely many directions gives a value above the true certified load factor, by up to about 7% with 1500 directions (Proposition 9.4). This is why the certificate is computed as a joint convex program rather than on a grid.

4.2 Field population

The earlier synthetic tests used Gaussian blobs. They are replaced by seven contact-mechanics families (`pfb.fields`, Figure 1), with three random fields per family on a unit pad: Hertz; the

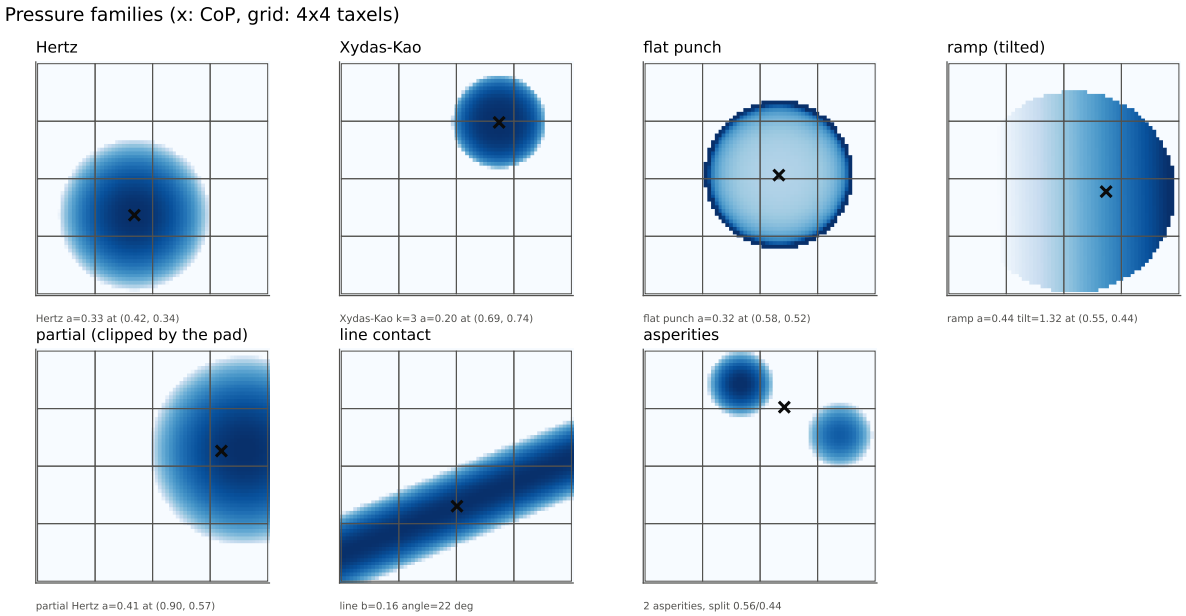


Figure 1: One field per family, with 4×4 taxels and the CoP (`examples/06_fields.py`).

soft-finger family of Xydas and Kao (1999) ($p \propto (1 - (r/a)^k)^{1/k}$); a rigid flat punch (pressure concentrated at the rim); a tilted ramp over a disk; a partial contact clipped by the pad edge; a line contact (cylinder); and two asperities. The last four lie outside the axisymmetric families assumed by parametric limit-surface models.

4.3 F1: guaranteed and possible wrench sets

Question. What do the three measurement models guarantee and allow on a contact outside the parametric families (Q2, Q3)?

Protocol. A partial, tilted contact (ramp clipped by the pad edge). The $(f_{T,1}, \tau)$ slice of the true set $\mathcal{X}(\nu_N)$ is compared with $\mathcal{X}^-$ and $\mathcal{X}^+$ for $\mathcal{M}_0$, $\mathcal{M}_F$, $\mathcal{M}_T$ at 4×4 and 16×16 taxels, with the torque about the true CoP for all three models (`examples/07_wrench_sets.py`).

Result. Figure 2. The $\mathcal{M}_0$ guaranteed set is the force segment: no torsion about the CoP is guaranteed (Corollary 4.3), and its possible set is wide. Taxel measurements give guaranteed sets with nonzero torsion, which close in on the true set at 16×16. $\mathcal{M}_T$ is much tighter than $\mathcal{M}_F$: mean widths 0.129 ($\mathcal{M}_F$) and 0.039 ($\mathcal{M}_T$) at 4×4, 0.032 and 0.009 at 16×16 (0.107 for $\mathcal{M}_0$).

Interpretation. An F/T sensor cannot support a torsional grasp decision, however accurate it is; distributed sensing is necessary, and resolution decides how much of the true capacity can be used.

4.4 F2: resolution and the value of measuring the CoP

Question. How do the bounds tighten with resolution, and what does the CoP add (Q2, Q3)?

Protocol. Population of 21 fields, $k \times k$ taxels with $k = 1, 2, 4, \ldots, 32$ (a dyadic chain, so the bounds tighten monotonically, Proposition 8.2), two twists about the true CoP of each field: the unit rotation $q = (0, 0, 1)$ and a mixed twist $q \propto (1, 0, 1/0.2)$ that balances force and torque at a

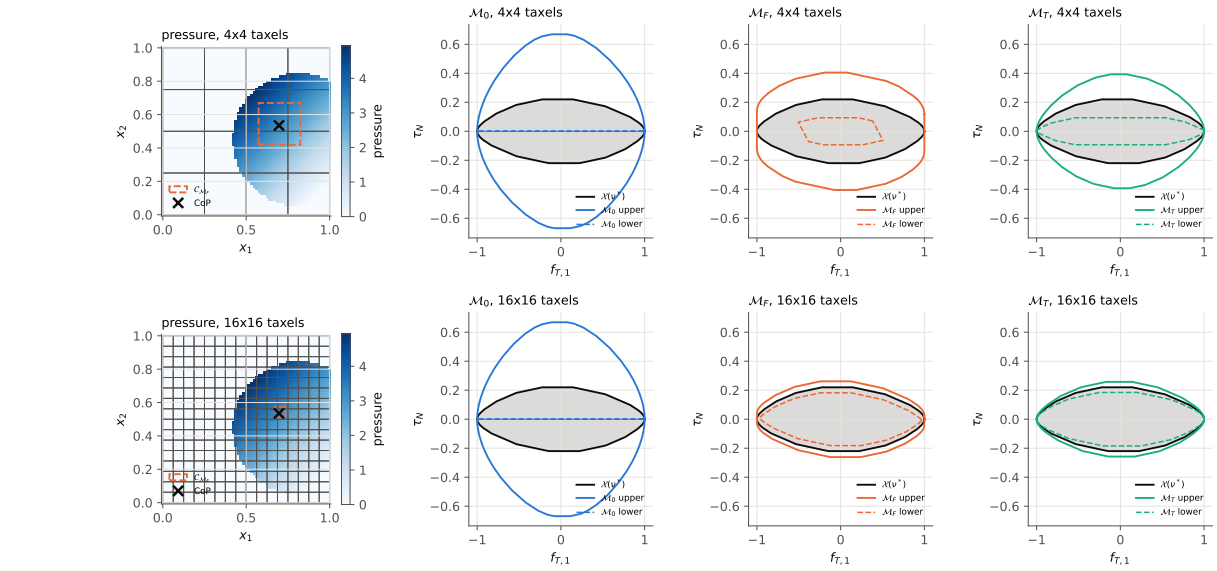


Figure 2: F1. True $\mathcal{X}(\nu_N)$ (grey), possible set $\mathcal{X}^+$ (solid) and guaranteed set $\mathcal{X}^-$ (dashed) for $\mathcal{M}_0$, $\mathcal{M}_F$, $\mathcal{M}_T$ on a partial tilted contact, torque about the true CoP; rows: 4×4 and 16×16 taxels.

lever of 0.2 pad sides (`examples/08_resolution.py`). All bounds are exact. The gap is compared with Proposition 8.3 and Theorem 8.4.

Result. Table 2 and Figure 3.

- For the rotation about the CoP, the median $\mathcal{M}_T$ lower bound rises from 0 (one taxel: Corollary 4.3) to 0.30 of the true support at 2×2, 0.68 at 8×8, 0.83 at 16×16 and 0.92 at 32×32; the upper bound falls from about 3 to 1.09.
- For the mixed twist, $\mathcal{M}_T$ guarantees 0.72 already with one taxel, and 0.96 at 32×32. $\mathcal{M}_F$ guarantees nothing with one taxel and only 0.31 at 2×2: without the CoP, the center of rotation of the twist may lie under the load.
- The gap $(\overline{h} - \underline{h})/(\mu |b| f_N)$ decreases linearly with the maximal taxel diameter and stays below the bound of Proposition 8.3, which holds for both models. From 4×4 on, each refinement divides the median $\mathcal{M}_F$ gap by 1.97–2.00 (largest ratio over the fields 0.57, 0.53 and 0.505 for the three steps), as predicted by Theorem 8.4, and the median $\mathcal{M}_F$ gap is 0.86–0.90 of the bound, inside the predicted range $[1/\sqrt{2}, 1]$. The $\mathcal{M}_T$ gap shrinks at the same rate (median ratios 0.47–0.52).
- For the rotation about the CoP, measuring the CoP narrows the bounds by 12% at 2×2 and by 2–8% from 4×4 on. For the mixed twist it narrows them by 30–34% at every resolution from 2×2 to 32×32 (46% with one taxel).

Interpretation. Resolution is what makes torsion certifiable: from a guaranteed zero with an F/T sensor to more than 80% of the true torsional support at 16×16. The rate is the one predicted by Theorem 8.4: each halving of the taxel size halves the gap, and the bound of Proposition 8.3 is attained up to the factor of the theorem, so it is a usable design rule for sensor resolution. The value of the CoP depends on the twist. For a rotation about the CoP itself, the taxel forces

	rotation about the CoP, $q = (0, 0, 1)$			mixed $q \propto (1, 0, 1/0.2)$ about the CoP		
taxels	$\mathcal{M}_T$ lower / upper	$\mathcal{M}_F$ lower / upper	CoP gain	$\mathcal{M}_T$ lower / upper	$\mathcal{M}_F$ lower / upper	CoP gain
1×1	0.00 / 2.97	0.00 / 3.94	23%	0.72 / 2.33	0.00 / 3.13	46%
2×2	0.30 / 2.48	0.18 / 2.74	12%	0.75 / 1.99	0.31 / 2.28	31%
4×4	0.48 / 1.78	0.48 / 1.81	3%	0.78 / 1.52	0.54 / 1.62	30%
8×8	0.68 / 1.39	0.68 / 1.39	2%	0.85 / 1.23	0.73 / 1.30	31%
16×16	0.83 / 1.19	0.83 / 1.19	2%	0.92 / 1.11	0.86 / 1.14	31%
32×32	0.92 / 1.09	0.91 / 1.09	8%	0.96 / 1.05	0.92 / 1.07	33%

Table 2: F2: median bound / true support over 21 fields (exact bounds). CoP gain: median of $(\text{width}_{\mathcal{M}_F} - \text{width}_{\mathcal{M}_T})/\text{width}_{\mathcal{M}_F}$.

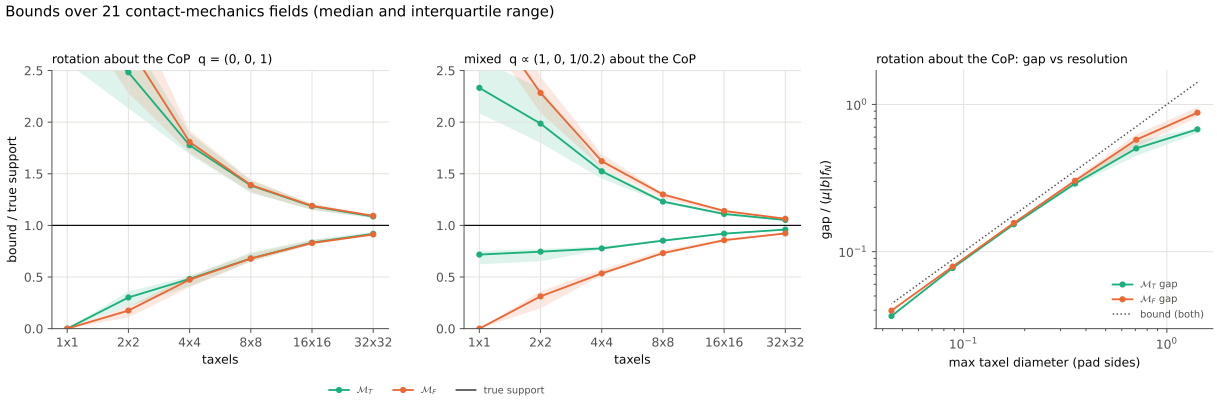


Figure 3: F2. Bound / true support against resolution for $\mathcal{M}_T$ and $\mathcal{M}_F$ (median and interquartile band over the population), and the gap against the maximal taxel diameter with the bound of Proposition 8.3 (both models).

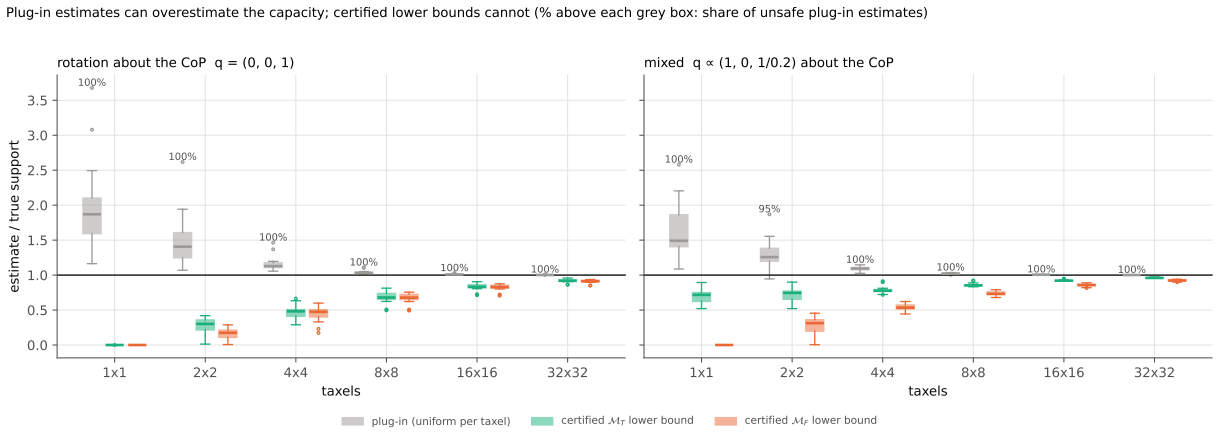


Figure 4: F3. Plug-in / true (grey) and certified lower bound / true ($\mathcal{M}_T$ green, $\mathcal{M}_F$ orange) against resolution; the percentage is the share of fields where the plug-in overestimates.

already locate the load well, and the CoP adds little beyond 2×2. For a twist that combines force and torque, the CoP fixes the lever arm, and measuring it removes about a third of the width at every resolution.

Changes from the previous version of this report. The previous version took torques about the CoP of each candidate measure. Its $\mathcal{M}_T$ and $\mathcal{M}_0$ values are unchanged (they are the same quantities, by Lemma 3.4), but its $\mathcal{M}_F$ columns bounded a twist that moved with each candidate's CoP; they were tighter for the mixed twist (lower bound 0.72 at one taxel) and suggested that the CoP adds little at intermediate resolution. That formulation is no longer used (Remark **??**). The previous study also used a 65×65 support grid; its 25–30% CoP gain at 32×32 was largely an effect of that grid (2 grid spacings per taxel), which the 129×129 grid removes.

4.5 F3: certification versus plug-in estimation

Question. Is a plug-in estimate safe (Q4)?

Protocol. Same population and directions (`examples/09_plugin.py`): plug-in / true, and certified lower bound / true for $\mathcal{M}_T$ and $\mathcal{M}_F$.

Result. Figure 4. The plug-in support value, evaluated at the same twist, exceeds the truth for every field at every resolution for the rotation about the CoP, and for all but one field (95%) at 2×2 for the mixed twist. The worst overestimate for the rotation is $\times 2.6$ at 2×2, $\times 1.47$ at 4×4, $\times 1.11$ at 8×8 and $\times 1.03$ at 16×16 (mixed twist: $\times 1.87$, 1.15, 1.04, 1.01). The certified lower bounds are below the truth by construction (medians as in Table 2).

Interpretation. The integrand g is convex in x (Lemma 3.12), so load spread out carries more torsion than the same load concentrated at its centroid. A plug-in estimate spreads each taxel force over the whole cell, including parts the real contact does not touch (the contact edge, partial contacts), and is therefore biased toward overestimation, most strongly in torsion and at coarse resolution. The bias shrinks with resolution and changed sign in one case only, so a plug-in estimate needs a margin tuned on data; the certificate needs none. This is the comparison the closed loop repeats (Section 5.4).

	pure torque $w = (0, 0, 1)$				force at lever 0.2, $w = (1, 0, 0.2)$			
taxels	$\mathcal{M}_T$	$\mathcal{M}_F$	$\mathcal{M}_T$ upper	plug-in (share > 1)	$\mathcal{M}_T$	$\mathcal{M}_F$	$\mathcal{M}_T$ upper	plug-in (share > 1)
1×1	0.00	0.00	2.93	1.75 (100%)	0.00	0.00	1.34	1.22 (81%)
2×2	0.24	0.00	2.46	1.39 (100%)	0.39	0.00	1.33	1.15 (95%)
4×4	0.46	0.43	1.79	1.14 (100%)	0.61	0.47	1.21	1.06 (95%)
8×8	0.68	0.68	1.38	1.04 (100%)	0.80	0.75	1.14	1.02 (95%)
16×16	0.83	0.83	1.19	1.01 (100%)	0.91	0.87	1.08	1.01 (95%)
32×32	0.92	0.91	1.09	1.00 (100%)	0.96	0.93	1.04	1.00 (95%)

Table 3: Load factors: median certified / true and upper / true load factor ($\mathcal{M}_T$; $\mathcal{M}_F$ upper: 3.23, 2.46, 1.80, 1.38, 1.19, 1.09 for the torque and 1.44, 1.44, 1.37, 1.23, 1.12, 1.06 for the lever load), and plug-in / true over 21 fields, with the share of fields where the plug-in overestimates. Solve time of one certificate (one pad): 9–11 ms up to 4×4, 12–13 ms at 8×8, 22–24 ms at 16×16, 74–77 ms at 32×32.

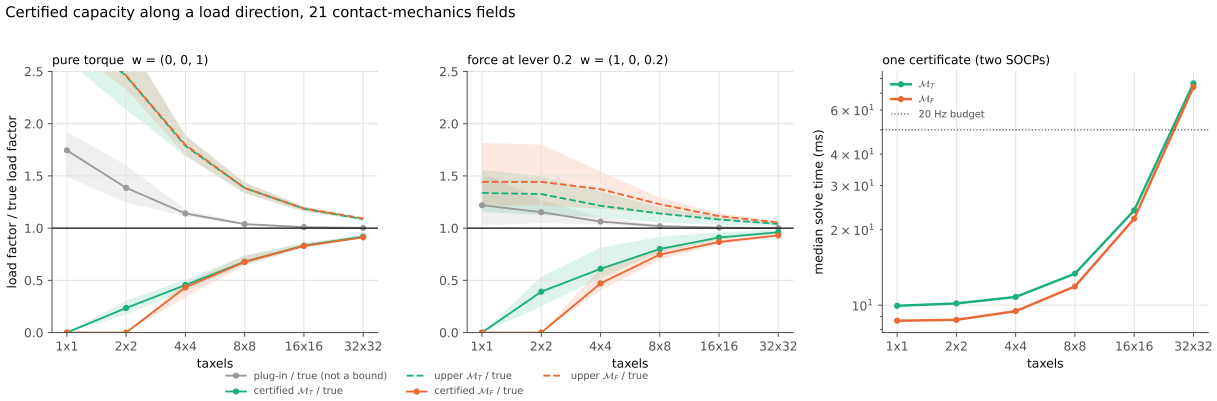


Figure 5: Certified (solid) and upper (dashed) load factor / true load factor for $\mathcal{M}_T$ and $\mathcal{M}_F$ against resolution (median and interquartile range), plug-in / true, and solve time of one certificate.

4.6 Certified load factor and runtime

Question. Support values describe the sets; a controller needs the capacity along a load. How tight and how fast are the certified and upper load factors (Q2–Q4)?

Protocol. Same population; loads w about the pad centre: a pure torque and a force at lever 0.2; $\underline{s}/s^\star$ and $\overline{s}/s^\star$ for $\mathcal{M}_T$ and $\mathcal{M}_F$, the plug-in load factor, and the solve time of one certificate (`examples/10_load_factor.py`).

Result. Table 3 and Figure 5.

Interpretation. The certified load factor behaves like the support bounds: from zero guaranteed torque with an F/T sensor to 83% of the true torque capacity at 16×16, within tens of milliseconds, fast enough for a 20 Hz control loop. With a force at a lever arm the one-taxel certificate is also zero, because the load contains a torque about o. Without the CoP ($\mathcal{M}_F$) the certificate is zero at 2×2 and lags $\mathcal{M}_T$ by up to 14 points for loads with a force component, where the unknown CoP shifts the lever arm. The plug-in load factor overestimates in all fields for a pure torque and in 81–95% for the lever load. The values are those of the previous version of this report up to the finer support grid (the certified load factor did not change with the fixed-frame formulation, which it already used). Pure-torque capacity is $\inf_a h((a, 1))$, slightly below the support value

$h((0,0,1))$ of F2, which is why the load-factor ratios at 2×2 (0.24) are below the support ratios (0.30).

Decision band. The two factors bracket the true capacity: a load is surely held below $\underline{s}$ and surely lost above $\overline{s}$ (Corollary 9.3). For a pure torque with $\mathcal{M}_T$, the undecided band shrinks from $[0.24, 2.46]$ times the true capacity at 2×2 to $[0.83, 1.19]$ at 16×16 and $[0.92, 1.09]$ at 32×32. A controller can therefore also detect, from the measurement alone, that a demanded load exceeds what any compatible contact could carry: for the median field at 16×16, any pure torque more than 19% above the true capacity. In the closed loop of S3 and S4 only the certified factor was used.

5 Drake simulation studies

5.1 Why Drake, and what it can test

Drake's hydroelastic contact computes a contact surface with a pressure field; in its discrete solvers each face of that surface is a compliant point contact at its centroid, with force $A_f p_f$ and Coulomb friction, and torsion emerges from the distributed points (Masterjohn et al., 2022). Two consequences:

- For frozen normal forces, the static capacity of a simulated contact *is* the support function of the theory on a discrete field. The true capacity of every simulated state is therefore known exactly, and a pressure field is available at every step, which hardware cannot provide.
- The dynamics are not those of the theory: compliance, regularized friction, time stepping, actuator lag and load dynamics. These are what the simulation tests, together with the sensing pipeline.

Drake cannot validate Coulomb friction on a gel, and a slip in simulation is a slip of Coulomb's law as Drake regularizes it. Tactile simulators were considered and not used: the mechanics-based ones also implement Coulomb friction, so checking slip against the bounds would be circular, and none would test the silicone behaviour reported by Howe and Cutkosky (1996, §3). MuJoCo was excluded because its torsional friction is a lumped coefficient with no pressure field.

Sensing. A pad's measurement is read from the contact surface: each face is one point load (pressure × area at its centroid); taxel forces are the sums over the faces whose centroids fall in each taxel of a $k \times k$ grid on a 32×32 mm footprint centred at the reference point o; the CoP is the force-weighted mean. Faces on non-parallel side walls are excluded and monitored; states with more than 0.5% of the force on side walls or outside the footprint are rejected. All wrenches are expressed in the grasp plane in common coordinates $u = (z, x)$, torques about the grip axis y, reference $o = 0$, loads $w = (f_z, f_x, \tau_y)$. The certificates of S2–S4 were always computed about this fixed point, so the fixed-frame formulation of the theory leaves S1–S4 unchanged.

5.2 S1: contact read-out

Question. Does the read-out of Drake's contact surface give the forces Drake applies (Q1)?

Protocol. A rigid block resting on a compliant pad (`sim/smoke_contact.py`).

Result. The sum of the face forces is 4.8957 N, against Drake's resultant 4.9045 N and the weight 4.9050 N (98 faces); the CoP is within 0.03 mm of the block's placement. The true pure-torque capacity of the contact (17.480 N mm) is close to its support value $h(0,0,1) = 17.485$ N mm, and to the capacity of a uniform 12×12 mm square (18.016 N mm).

Setting	$s_{\text{obs}}/s_{\text{true}}$	certified $/s_{\text{obs}}$
lagged, stiction tolerance 10^{-6} m/s (reference)	1.001	0.915
time step $2.5 \cdot 10^{-4}$ / ramp 0.5 N/s / mesh 0.5 mm / threshold 0.3 mm/s / modulus $2 \cdot 10^{6}$	0.992–1.001	0.915–0.923
stiction tolerance 10^{-5}	0.997	0.918
stiction tolerance 10^{-4} (Drake default)	**0.893**	**1.025** (violated)
similar, stiction 10^{-6}	1.001	0.915
SAP (automatic regularization)	**1.360**	0.673

Table 4: S2: break-away of the bar against the true and certified load factors, over solver settings. Side walls carried 0.00% of the force.

Interpretation. The face read-out is a faithful measurement of the simulated contact (0.2% below Drake's resultant), so the taxel forces of the later studies are those of the simulated pressure field.

5.3 S2: break-away against the certificate, and solver sensitivity

Question. Does a simulated grasp break away at its true load factor, and does the certified load factor stay below the observed break-away (Q1)? Is the answer robust to the simulator's settings?

Protocol. A rigid bar squeezed between two flat compliant pads (20×16 mm, about 7.9 N each, no gravity). A force along z is ramped slowly at a 30 mm lever arm in the mid-plane, so the load direction $w = (1, 0, -0.03)$ combines shear and torque about the grip axis. The observed break-away s_{obs} is the force at which the bar speed exceeds 1 mm/s; s_{true} is the exact load factor of Drake's faces just before slip; the certificate is $\mathcal{M}_T$ 16×16 on both pads. The sweep varies the discrete contact solver, its stiction tolerance, the time step, ramp rate, mesh size, detection threshold and pad modulus (`sim/sweep_breakaway.py`).

Result. Table 4.

Interpretation.

- With the lagged or similar solvers and a stiction tolerance of at most 10^{-5} m/s, the bar breaks away within 1% of the ideal Coulomb capacity, insensitive to time step, ramp rate, mesh and stiffness, and the certificate stays below it (0.915–0.923 at 16×16; about 0.5 with one pad unsensed). Two-pad additivity (Proposition 9.6) holds in this setting.
- Drake's default tolerance makes points near the rotation centre creep below full friction, so the bar breaks away at 89% of its capacity and a sound certificate *appears* violated. SAP holds 36% above the Coulomb capacity although its normal forces match the pressure field within 0.6%; the cause is unexplained and SAP is excluded.
- A simulator's friction regularization can contradict a Coulomb guarantee. S3 and S4 use the lagged solver with stiction tolerance 10^{-6}, and the setting must be reported with any simulation result.
- Runtime: a two-pad $\mathcal{M}_T$ 16×16 certificate takes about 41 ms serially (one pad: 22 ms), within a 20 Hz control budget.

5.4 S3: closed-loop certified grip control

Question. If a controller squeezes just enough to keep the demanded load certified, does the grasp hold, at what squeeze, and how does this compare with nominal estimators whose margins are tuned on data (Q5)?

Scene. A rigid bar ($150 \times 20 \times 60$ mm, gravity off) moves on a planar joint (translations x, z, rotation about the grip axis y) between two compliant pads (hydroelastic modulus $5 \cdot 10^5$ Pa, $\mu = 0.8$), each on a prismatic joint and pushed with the same squeeze, which follows the command through a first-order filter ($\tau = 50$ ms, the gripper bandwidth). Three pad families, both pads drawn independently within the family: *dome* (ellipsoid, Hertz-like field whose area grows with the squeeze), *tilted* box (ramp field or partial edge contact) and *flat* box (near-uniform field, the control case). Solver: lagged, stiction tolerance 10^{-6}.

Loads. The demanded grasp wrench is $D(t)\, w(t)$ in the grasp plane. 20% of the seeds demand a pure torque, the others a force at a lever of 2–6 RMS contact radii, so that torsion binds (the peak squeeze is at least three times the shear-only value). Scenario A (the hardware task): D ramps to $D_{\max}$ over 3 s along a fixed w, then holds for 1 s. Scenario B: D oscillates between 0.3 and 1 times $D_{\max}$ while the lever swings between 0 and its maximum over 7 s, so the binding direction changes. 30 seeds per scenario.

Controller (identical for every method). Every 50 ms the method estimates the load factor $\hat{s}$ of the current contact along w and commands

$$f_{\text{cmd}} = \text{clip}\left(f_{\text{meas}} \frac{D\,(1+m)}{\hat{s}},\ F_{\min},\ f_{\max} \right), \tag{1}$$

with $F_{\min} = 1$ N, decreases limited to 2% per cycle, and D, w taken 150 ms ahead (one control period plus about two filter time constants; the load profile is known). In B, the larger of the commands for the current and the upcoming load is used. Three declared heuristics lie outside the theory: (i) capacity scales linearly with squeeze ($\hat{s}/f_{\text{meas}}$ is a capacity per newton; on domes it grows with squeeze, so the update is conservative there); (ii) the relaxation limit; (iii) the load preview. A certificate is sound for the measured state only.

Force budget and flag. Pushing a thin compliant pad past about half its thickness flattens and then reverses the hydroelastic force–depth law. In a pilot, the coarse 2×2 certificate at $m = 1$ commanded 46 N on a dome while the measured squeeze fell from 44.6 to 25 N, and the bar slipped. The command is therefore clipped at a per-seed budget $f_{\max}$, the static squeeze at 50% of the pad's semi-thickness (18–76 N), and an episode is *flagged* at the first cycle where the requested squeeze exceeds it.

Free object. In a pilot with a free bar, pads whose normal-force centres did not line up created a couple about an in-plane axis that opposite tangential forces at the two faces had to balance; the oracle slipped at a 5% margin with a bar tilt of about 1.6°. The planar certificate does not account for this internal load: Proposition 9.6 assumes the out-of-plane components are balanced without friction. The planar joint enforces that assumption in the study; with a real free object it is a limitation (Section 9, scope).

Design. 30 seeds $\times$ 2 scenarios $\times$ (3 oracle margins + 10 methods $\times$ 6 margins, $m \in \{0, 0.05, 0.1, 0.25, 0.5, 1\}$) = 3780 episodes (1.8 h on 22 processes); none was rejected by the side-wall and footprint filters.

Result 1: outcomes. Table 5 (scenario A; B is within a few points) and Figure 6.

- **No certified silent slip:** 0 of 2160 certified episodes, at every margin, in both scenarios and all pad families; the largest slip was 0.08 mm against the 0.2 mm threshold. 228 certified episodes were flagged; none slipped afterwards. Since the certificate is conservative, a flag means "cannot certify within the budget", not "will slip".

Method	$m = 0$	$m = 0.05$ (m_{dyn})	$m = 0.25$	$m = 1$
oracle ($m = 0$ / 0.05 / 0.1)	3/0/97 · 0.96	100/0/0 · 1.00	(0.1) 100/0/0 · 1.05	
certified $\mathcal{M}_T$ 2 × 2	87/13/0 · 2.95	83/17/0 · 3.10	70/30/0 · 3.69	7/93/0 · 3.85
certified $\mathcal{M}_T$ 4 × 4	93/7/0 · 2.37	93/7/0 · 2.48	93/7/0 · 2.94	47/53/0 · 5.87
certified $\mathcal{M}_T$ 8 × 8	100/0/0 · 1.33	100/0/0 · 1.40	100/0/0 · 1.66	93/7/0 · 2.63
certified $\mathcal{M}_T$ 16 × 16	100/0/0 · 1.13	**100/0/0 · 1.18**	100/0/0 · 1.40	100/0/0 · 2.20
certified $\mathcal{M}_F$ 16 × 16	100/0/0 · 1.17	100/0/0 · 1.23	100/0/0 · 1.46	100/0/0 · 2.26
certified, one pad sensed	100/0/0 · 2.30	97/3/0 · 2.40	83/17/0 · 2.80	60/40/0 · 4.72
plug-in 4 × 4	3/0/97 · 0.97	13/0/87 · 0.97	60/0/40 · 1.11	97/0/3 · 1.70
plug-in 16 × 16	23/0/77 · 0.96	77/0/23 · 0.99	100/0/0 · 1.16	100/0/0 · 1.85
F/T + contact mask	0/0/100	50/0/50 · 0.99	100/0/0 · 1.16	100/0/0 · 1.83
uniform over pad (F/T only)	0/0/100	0/0/100	0/0/100	73/0/27 · 1.20

Table 5: S3, scenario A: success / infeasible / silent slip (%) · median effort of the successes (squeeze / oracle at m_{dyn}).

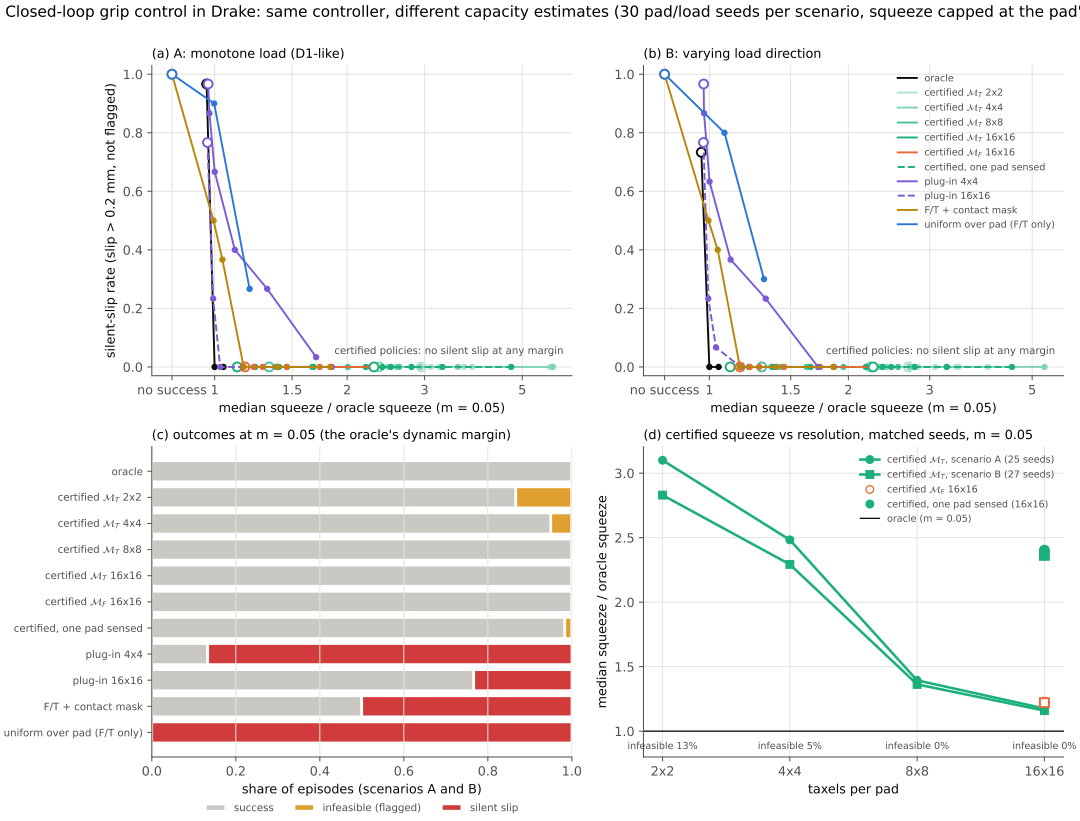


Figure 6: S3. (a, b) Silent-slip rate against median squeeze for every method and margin, scenarios A and B. (c) Outcomes at m_{dyn}. (d) Certified squeeze against resolution on matched seeds, with the infeasible rate.

	2×2	4×4	8×8	16×16	$\mathcal{M}_F$ 16×16	one pad
A (25 seeds)	3.10	2.48	1.39	1.18	1.23	2.40
B (27 seeds)	2.83	2.29	1.36	1.16	1.22	2.36

Table 6: S3: median squeeze / oracle at m_{dyn} on the seeds where the oracle and all certified methods succeed.

Method	tuned on A seeds 0–14 → test A 15–29; B	tuned on flat pads → test curved pads A; B
plug-in 4×4	no safe margin up to $m = 1$	$m = 0.1 \rightarrow 100\%$; 95%
plug-in 16×16	$m = 0.1 \rightarrow 0\%$ (1.03 vs 1.20); 7% (1.03 vs 1.16)	$m = 0.05 \rightarrow 35\%$; 35%
F/T + contact mask	$m = 0.25 \rightarrow 0\%$ (1.11 vs 1.20); 0% (1.16 vs 1.16)	$m = 0.05 \rightarrow 75\%$; 75%
uniform over pad	no safe margin up to $m = 1$	$m = 1 \rightarrow 40\%$; 45%

Table 7: S3: silent-slip rate on the test set at the tuned margin, and effort against certified $\mathcal{M}_T$ 16×16 at m_{dyn} (matched seeds).

- The nominal methods slip *silently* whenever they slip: their estimate never asks for more than the budget.
- **Infeasibility is the cost of coarse resolution:** at m_{dyn}, 13% (2×2), 5% (4×4) and 0% (8×8, 16×16) of the episodes, pooled over A and B. The flagged seeds have budgets from 18 to 76 N, so the flags come from a high certified demand (large lever arms on coarse taxels), not only from small budgets.
- **F/T only fails** even at $m = 1$ (27% silent slips): the real contacts are smaller than the pad.

Result 2: effort on matched seeds. Table 6.

Resolution buys most of the efficiency: the certified squeeze falls from about 3 times the oracle at 2×2 to 1.18 at 16×16. Measuring the CoP adds little at 16×16 ($\mathcal{M}_F$ 1.23). Sensing one pad only costs as much as dropping to about 4×4 on both pads (2.40): the unsensed pad's unknown pressure is charged at its worst case.

Result 3: nominal margins, tuned then tested. The margin of each nominal method is tuned as the smallest margin without silent slip on a training set, then tested on held-out data, with two splits: by seed (tune on A seeds 0–14, test on A seeds 15–29 and on B), and by contact geometry (tune on the flat pads of A, test on the curved pads, dome and tilted, of A and B). Table 7.

- **On a representative training set, tuned nominal methods are cheaper.** Plug-in 16×16 at $m = 0.1$ needs 1.03 times the oracle squeeze against 1.16–1.20 for the certificate, but slips in 7% of B; the contact-mask method at $m = 0.25$ needs 1.11–1.16 and transfers from A to B without slip. The certificate costs 0–17% more squeeze than the best tuned baseline (equivalently, the baseline needs 0–14% less).
- **Under a shift of contact geometry, tuned margins fail.** Margins tuned on flat pads slip in 35% (plug-in 16×16) to 75% (contact mask) of the curved-pad episodes. The contact-mask method fails exactly where its uniform-pressure assumption is wrong: 100% silent slip on domes at m_{dyn}, 0% on flat pads.

Result 4: the oracle at zero margin. The oracle slips at $m = 0$ (97% in A, 73% in B). The slips occur in the hold phase after the ramp (median 3.05 s, the ramp ends at 3 s), where the loop converges to capacity / demand ≈ 1.01; during the ramp the preview keeps it near 1.04. At

$m = 0.02$, the oracle holds with 0.06 mm of creep. A zero-margin decision therefore leaves no room for model error (the simulated break-away matched the computed capacity only to 0.992–1.001 in S2): this is a boundary effect, not evidence that static admissibility is insufficient.

Interpretation.

- The certificate holds in closed loop, including on contacts far from any parametric family, and signals infeasibility before slip. The loop needs about 5% of slack (the oracle's dynamic margin), which a certificate at m_{dyn} provides; the zero-failure result at $m = 0$ of the certified methods relies on their conservatism and is not a dynamic guarantee.
- The value of certification is *robustness, not efficiency*: no calibration set, an explicit infeasibility flag, and immunity to shifts of contact geometry. Its price at 16×16 is 16–18% squeeze over the oracle and 0–17% over the best tuned baseline.
- Decision time under the 22-process pool was 56–172 ms for the certificates; serially, on frozen states, 14–45 ms (two-pad $\mathcal{M}_T$ 16×16 about 40 ms), within 20 Hz.

5.5 S4: sensing delay and load rate

Question. The certificate concerns the measured state. How much sensing delay does the S3 loop tolerate, with the S3 margins and no retuning (Q6)?

Where the delay acts. Under linear scaling, $\hat{s}/f_{\mathrm{meas}}$ in the S3 command is a capacity per newton. It depends on the shape of the pressure field, not on the squeeze level, and changes slowly, so a stale tactile state should matter little. The load is different: S3 knew the load profile 150 ms ahead, whereas in the hardware task (a cup being filled) the load is measured, late. The study therefore runs the same controller in two information modes:

- **P (load previewed):** the S3 loop; the tactile state and f_{meas} are Δ old.
- **M (load measured):** the load is also measured Δ late, with no preview: $D(t-\Delta)$ replaces $D(t + 150\,\mathrm{ms})$.

Design. Scenario A only (the hardware task); delays $\Delta \in \{0, 25, 50, 100, 200\}$ ms (25–50 ms: a 30–60 fps vision-based tactile frame plus processing; 100–200 ms: a slow pressure reconstruction); load-rate factor 1, 2, 4 (ramp 3, 1.5, 0.75 s); six methods at their S3 margins: oracle, $\mathcal{M}_T$ 8×8, $\mathcal{M}_T$ 16×16 and one pad at $m_{\mathrm{dyn}} = 0.05$, plug-in 16×16 at its tuned 0.1, contact mask at its tuned 0.25. 30 seeds; 5400 episodes, none rejected. The delayed state is sampled at the simulation's 5 ms sub-steps. With $\Delta = 0$ and preview, the episodes reproduce S3 exactly. Not varied, with reasons: the control rate (a held command acts like extra delay, the 16×16 two-pad certificate needs about 40 ms so 50 Hz is unrealistic, and changing the period also changes the preview horizon); a slower load (easier than nominal); scenario B. **Prediction stated before the run:** P flat; M degrading for every method, the oracle included.

Slip-onset diagnosis. At the first step where the slip exceeds 0.05 mm, the shortfall of the true capacity s_{true} of the current contact against the applied demand D_{now} splits exactly against the last decision (sensed state, load D certified, estimate $\hat{s}$, command f_{cmd}):

$$\ln \frac{D_{\mathrm{now}}}{s_{\mathrm{true,now}}} = \underbrace{\ln \frac{\hat{s}}{s_{\mathrm{true}}(\mathrm{sensed})}}_{\text{estimate}} + \underbrace{\ln \frac{D_{\mathrm{now}}}{D}}_{\text{load}} + \underbrace{\ln \frac{\kappa_{\mathrm{sensed}}}{\kappa_{\mathrm{now}}}}_{\text{state}} + \underbrace{\ln \frac{f_{\mathrm{cmd}}}{f_{\mathrm{now}}}}_{\text{actuator}} + \underbrace{\ln \frac{\mathrm{target}}{f_{\mathrm{cmd}}}}_{\text{slack} \le 0} - \ln(1+m), \qquad (2)$$

with $\kappa = s_{\mathrm{true}}/f$ the true capacity per newton and the slack due to the squeeze floor or the relaxation limit. The cause of a slip is the largest of the four cause terms, or *model* if the

Method (margin)	1×: 0 / 25 / 50 / 100 / 200 ms	2×	4×	silent-free up to
oracle (0.05)	100 at every delay	100	100	none
$\mathcal{M}_T$ 8 × 8 (0.05)	20 / 20 / 23 / 67 / 90	20 / 67 / 67 / 97 / 100	67 / 73 / 97 / 100 / 100	none
$\mathcal{M}_T$ 16 × 16 (0.05)	23 / 60 / 67 / 73 / 100	67 / 67 / 87 / 100 / 100	70 / 97 / 100 / 100 / 100	none
one pad sensed (0.05)	0 / 0 / 0 / 0 / 47	0 / 0 / 0 / 47 / 100	0 / 33 / 57 / 100 / 100	100 / 50 / 0 ms
plug-in 16 × 16 (0.1)	100	100	100	none
contact mask (0.25)	40 / 100 / 100 / 100 / 100	100	100	none

Table 8: S4, mode M: silent-slip rate (%) against the sensing delay, per load rate; "silent-free up to" is the largest delay up to which no silent slip occurred, at rates 1× / 2× / 4×.

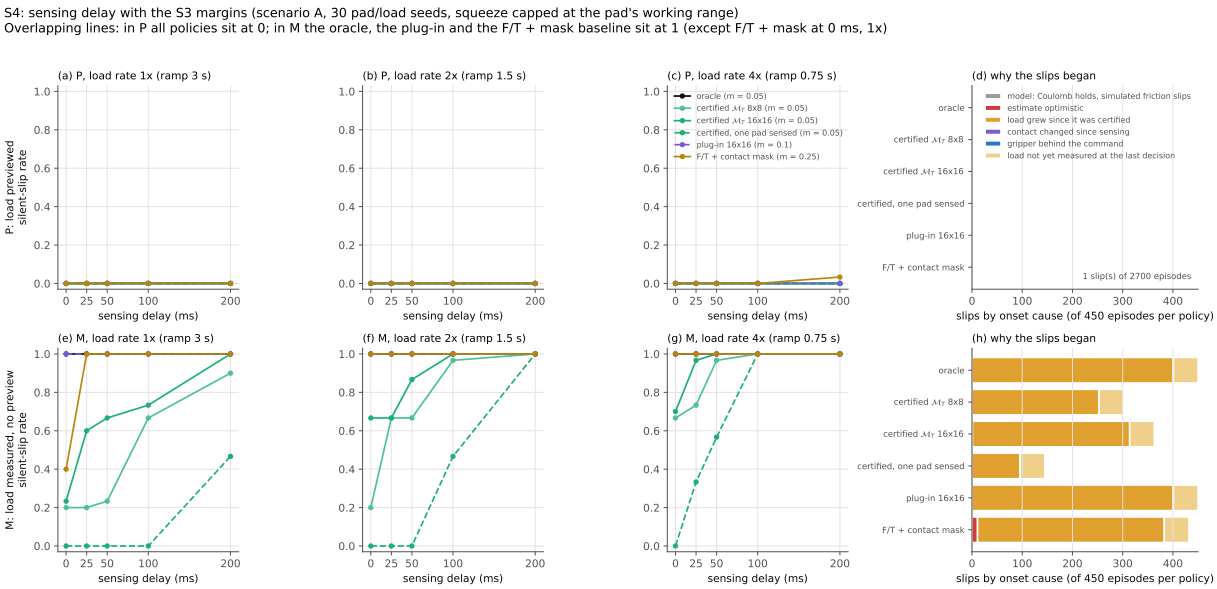


Figure 7: S4. Silent-slip rate against the sensing delay with the S3 margins; top row: load previewed (P), bottom row: load measured (M); columns: load rate 1×, 2×, 4×; right: onset cause of every slip.

Coulomb capacity still exceeds the demand ($s_{\text{true,now}} \geq D_{\text{now}}$), or *load not yet measured* if no load had been seen at the last decision. The identity was checked to round-off on test episodes.

Result: P (load previewed). No silent slip for any method, delay or rate, except one contact-mask seed at 4× and 200 ms (cause: estimate, +0.22). Effort does not change with the delay (oracle 1.00, $\mathcal{M}_T$ 8 × 8 1.40, 16 × 16 1.18, one pad 2.40–2.51, plug-in 1.03, mask 1.16); the one-pad certificate is flagged in 3% of the episodes (7% at 4× and ≥ 100 ms). A stale tactile state is harmless here; on domes the stale capacity per newton is the lower one, so the error is on the safe side.

Result: M (load measured). Table 8 and Figure 7.

- **Causes.** Of the 2142 slips in mode M, 1836 are *load* (the load grew since the one certified), 294 *load not yet measured*, 11 *estimate* (contact mask), 1 *model*, and none *state* or *actuator*. The median capacity / demand at onset is 0.75.
- **When.** The slips begin at load onset, at a fixed point of the ramp for every method and delay: 13% into the 3 s ramp, 20% into the 0.75 s ramp. They are real slips of 2–5 mm (median 2.6 mm), not boundary ticks. The episode in Figure 8 slipped early and then held; recovery was not checked systematically.

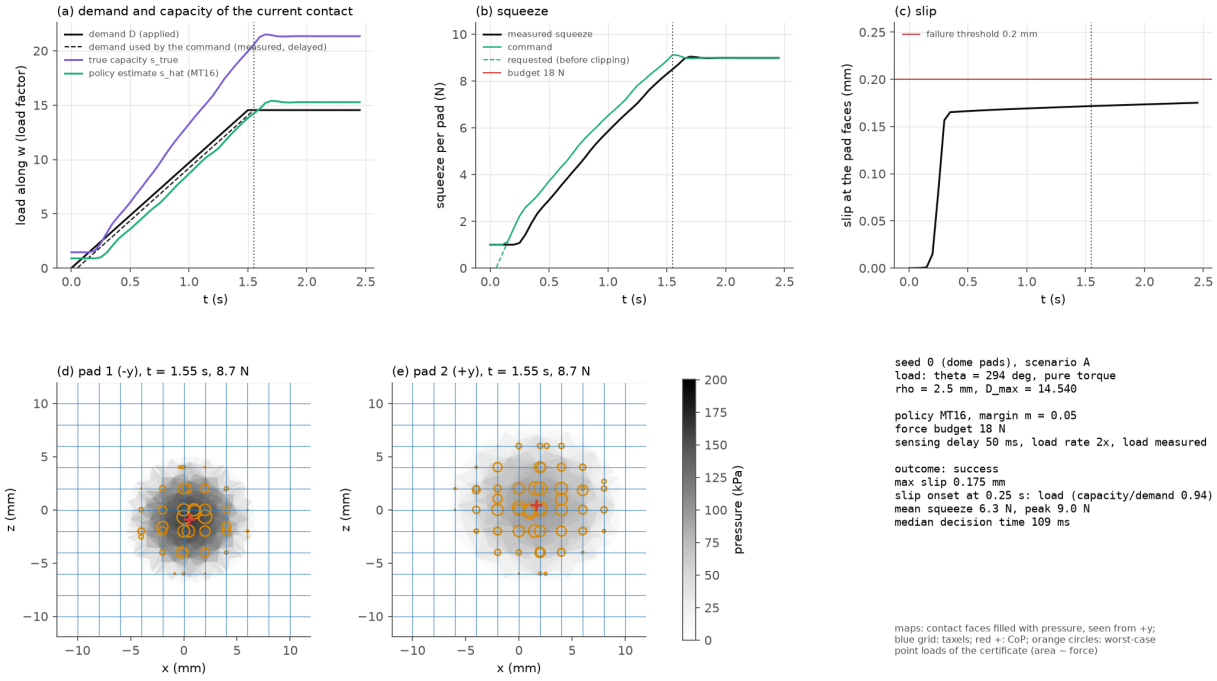


Figure 8: S4, one episode in mode M ($\mathcal{M}_T$ 16×16, seed 0, $\Delta = 50$ ms, rate $2\times$; `sim-show --delay 50 --rate 2 --no-preview`). (a) The measured demand lags the applied demand, and the true capacity dips below it at the start of the ramp; (c) the bar slips by 0.17 mm at load onset, just under the failure threshold, then holds once the squeeze catches up. Slip onset at 0.25 s, cause *load*, capacity / demand 0.94.

- **Why the oracle fails at $\Delta = 0$.** Without preview, the load keeps growing during the 50 ms command hold and the gripper lag. Early in a ramp from zero this growth is large in relative terms, about one control period divided by the elapsed time per cycle (13% at 0.39 s), which exceeds the multiplicative margin $m = 0.05$.
- **Certificates survive in proportion to their conservatism.** The median estimate term at onset is -0.88 (one pad), -0.30 (8×8), -0.15 (16×16) and about 0 (oracle, plug-in): the order of survival.

Interpretation.

- **The weak link is temporal, not the certificate**, as predicted. Delaying the tactile state costs nothing; not knowing the load ahead breaks every method, the oracle first.
- **The dynamic margin calibrated with preview does not transfer to a measured load.** $m_{\mathrm{dyn}} = 0.05$ covered the loop with preview only.
- **Conservatism acts as an implicit temporal margin.** This is conservatism, not a merit of certification: the guarantee concerns the sensed state, $w(t-\Delta) \in \mathcal{X}^{-}(y(t-\Delta))$, and says nothing about $w(t)$.
- **A static certificate needs a separate temporal term** covering the load change over the delay plus the loop lag. For a ramp from zero it cannot be a multiplicative margin, since the relative growth is unbounded at onset; it must be a load preview, an additive or rate-based term on the certified load (e.g. $D + \dot{D}\,L$ for a lag L), or a squeeze floor sized for the first cycles. None of these was implemented: the study stops at the requirement.

Claim	Evidence	Status
F/T + CoP never certifies torsion about the CoP	Cor. 4.3; F1; S3 (uniform over pad fails at $m = 1$)	proven; illustrated
Exact, tractable taxel bounds ($\mathcal{M}_F$ closed form, $\mathcal{M}_T$ SOCP/LP)	Thms 6.1, 7.1, 7.2; verification	proven; verified
Certified and upper load factors, tight, multi-contact; hold / slip / undecided	Prop. 9.2, Cor. 9.3, Props. 9.4–9.6; tests; load factor; S2	proven (sketch); verified
Gap halves per dyadic refinement; value of the CoP for mixed loads	Props. 8.1–8.3, Thm 8.4; F2; load factor; S3	proven; quantified
Plug-in estimates are unsafe	F3; load factor; S3	numerical, simulation
Certificate sound in closed loop; price	S3	simulation (Drake, planar joint)
Value is robustness, not efficiency	S3 tuned-margin transfer	simulation
Static certificate needs a temporal term	S4	simulation
Coulomb patch model holds on a gel	—	open (hardware)
Robust bounds under taxel noise	—	open (theory)

Table 9: Claims and the evidence in this report.

- This matches the observation of Costanzo et al. (2020, §V-C) that static limit-surface control alone slips under fast load changes, here with the mechanism isolated.

6 Synthesis

Limitations.

- Planar, quasi-static Coulomb friction with a known, uniform μ, and noise-free taxel forces.
- The closed-loop results use a planar joint for the bar (out-of-plane moments carried without friction) and Drake's lagged solver with a tight stiction tolerance; other settings change the outcome (S2).
- The controller's linear scaling of capacity with squeeze, its relaxation limit and the load preview are heuristics outside the theory.
- The dynamic margin is calibrated on the oracle in one loop; it is a property of the loop, not of the certificate.
- Simulated pads are hydroelastic: their pressure comes from penetration depth, not elasticity; they serve as test fields, not as a model of a gel.

7 Planned work

The outcomes of the following studies are open; this section states what they will test.

7.1 Robust extension

Motivation. Real taxel forces are noisy, and vision-based tactile sensors provide pressure only through an elastic reconstruction, so the measurement is an interval rather than a value. The friction coefficient is also uncertain.

Theory. Bounds and the certified load factor for interval taxel forces $F_i \in [F_i^-, F_i^+]$ (with consistent totals and, for $\mathcal{M}_T$, a convex set of CoPs) and $\mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$. The friction coefficient only scales the bounds. For a fixed twist, the $\mathcal{M}_T$ lower bound stays an SOCP with interval taxel forces, since it is jointly convex in the forces and the local first moments (Theorem 7.1). For the load factor, the twist is a variable and F_i multiplies the direction a, so the substitution $u_i = bc_i$ of Proposition 9.4 may not keep the problem jointly convex; another formulation or a relaxation with a certified gap may be needed.

Numerical study (F4: resolution against noise). Pixel noise aggregated into taxel intervals. The nominal gap shrinks like $1/k$ (Proposition 8.3) while the sum of the interval widths grows like $\sqrt{r}$ for $r = k^2$ independent taxels, so an interior optimal resolution is plausible; the study will test whether it exists and where.

Simulation. Repeat S3 with noisy taxels and the robust certificate, to test whether soundness survives and at what cost.

7.2 Hardware validation

Purpose. The only independent physical test: does the Coulomb patch model, which the certificate assumes, hold on a real sensor gel? Howe and Cutkosky (1996, §3) report that silicone and rubber fingertips slip below the Coulomb limit-surface prediction under combined torsion and shear, so soundness is not presumed; the study is a *model-adequacy test*, and violations will be reported.

Set-up. No gripper. A GelSight Mini is mounted on a six-axis ATI force/torque sensor at the robot wrist. The ATI gives the full contact wrench at 1 kHz, hence f_N and the CoP exactly (the $\mathcal{M}_0$ information). The GelSight supplies only the *shape* of the normal-force distribution. Objects touch the gel through small flat-topped bosses, so that the whole contact stays inside the field of view.

Calibration.

- **Pressure from depth.** The depth map is mapped to pressure inside the contact mask, integrated over $k \times k$ cells and scaled to the ATI's f_N. A local depth–pressure law is only a first approximation for a nearly incompressible gel, so the map is calibrated with statically determinate indenters: one to three bosses whose individual forces follow exactly from the ATI's force and moments. It is validated on held-out shapes against the ATI CoP. Its residuals give the taxel intervals of the robust extension.
- **Friction coefficient.** μ comes from pure-shear slides only, since translational capacity is μf_N for any pressure (Howe and Cutkosky, 1996, §2.3). Torsion and combined loads are held out as the test. The calibration records the dependence on normal force, speed and dwell time, and the drift.

Model-adequacy tests (single contact, indenter clamped to the table). T1: steady sliding at an imposed twist q, set by the instantaneous centre of rotation. Maximum dissipation gives $\langle w_{\text{meas}}, q \rangle = h(q)$, with the wrench and the twist written about the same point (the ATI reference), which is compared with $[\underline{h}(q), \overline{h}(q)]$ for $\mathcal{M}_0$, $\mathcal{M}_F$ and $\mathcal{M}_T$ at each resolution. T2: a tangential wrench ramped along a fixed w until gross slip, with s_{obs} compared with $[\underline{s}, \overline{s}]$, as in S2. The plug-ins, the contact-mask method and the parametric model of Costanzo et al. (2020) are evaluated on the same frames. Short slides with filleted bosses and an automatic abort protect the gel.

Demonstrations.

- **Dial (single contact).** The sensor turns a spring-loaded dial through friction, with the certified minimal press force under a pure torsional load that rises with the angle. This is a previewed load, mode P of S4.
- **D1.** A plate is held between the GelSight and a passive pad on a vertical bar while a container at a lever arm fills slowly. This is the hardware counterpart of S3/S4 scenario A, with the load either previewed (known flow) or measured.
  - Configuration V: the passive side has low friction, so the ATI measures the whole tangential load.
  - Configuration D: the passive side is a gel, with the load split measured as a side result.

Methods: F/T only, which cannot certify torsion; plug-ins; contact mask; parametric; certified at 2×2 to 16×16. Metrics: slip, flags, peak squeeze.

What the simulations imply for the design.

- With a measured load, the loop must certify the load growth over the measured latency (S4), or use a previewed load.
- A free object with offset normal-force centres loads friction out of plane (S3). Coaxial bosses and a load hung in the mid-plane avoid it.
- An unsensed contact is costly (2.4 times the oracle squeeze in S3). In configuration D its worst case is taken over the known boss face only.

A gate on recorded maps of the demo objects (certified gap of at least 1.5 between 2×2 and 16×16, forces within the gel's limit) precedes the demonstrations. The detailed protocol and phases are in `research/stage3_validation_plan.md`, §9.

A Reproducibility

Study	Command (`pixi run ...`)	Cache / output
Tests	`test`	—
Verification	`counterexample`, `bounds`, `refinement`	`examples/figures/`
Field population, F1	`fields`, `wrench-sets`	`doc/figures/fields.pdf`, `wrench_sets.pdf`
F2, F3	`resolution`, `plugin`	`examples/data/study.npz`
Load factor	`load-factor`	`doc/figures/load_factor.pdf`
S1	`-e sim sim-smoke`	`sim/out/smoke.*`
S2	`-e sim sim-sweep`	printed table
S3	`-e sim sim-grip`, `sim-grip-plot`	`sim/data/grip_study_v2.npz`
S4	`-e sim sim-delay`, `sim-delay-plot`	`sim/data/delay_study.npz`
One episode	`-e sim sim-show --seed 0 --policy MT16 [--delay 50 --rate 2 --no-preview]`	`sim/out/episode_*`

Table 10: Commands, run from the root of the code repository `pfb`; vector figures are written to `$PFB_DOC_FIG` (`make figures` in this repository sets it to `doc/figures/`). Drake runs in the pixi environment `sim`.

Settings: Drake discrete solver *lagged*, stiction tolerance 10^{-6} m/s, time step 1 ms, control period 50 ms, gripper filter 50 ms, load and filter update 5 ms, $\mu = 0.8$, slip threshold 0.2 mm at the pad faces. Seeds 0–29 cycle through the pad families (dome, tilted, flat). S3: 1.8 h, S4: about 1 h, on 22 processes.
Figures and numbers built against code commit `ca83b74-dirty`.

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